8

Jasmine Torres

CHE 313 Transport Phenomena III
Homework #2 (Due:2/05/26, 2:30 pm)
TA office hour (T,11am-1pm;TR, 12:30-2:30 pm): Aelx Acquah (aacqua2@uic.edu), EIB 170

Note: Explain your answers in detail!

1. ~~Textbook (Wankat, P. C. 5th edition), 2.D5~~
2. ~~Textbook (Wankat, P. C. 5th edition), 2.D10~~
3. ~~Textbook (Wankat, P. C. 5th edition), 2.D16.~~
4. ~~Textbook (Wankat, P. C. 5th edition), 2.D15. Please provide a detailed procedure for your solution.~~
5. ~~Textbook (Wankat, P. C. 5th edition), 2.D24. Please use a spreadsheet and submit it with your answer and graph.~~

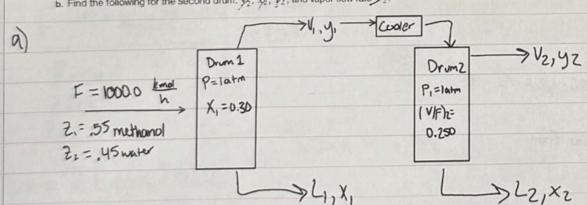
1.

2 D5. We have a feed that is a binary mixture of methanol and water (55.0 mol% methanol) that is sent to a system of two flash drums hooked together. The vapor from the first drum is cooled, which partially condenses the vapor, and then is fed to the second flash drum. Both drums operate at a pressure of 1.0 atm and are adiabatic. The feed rate to the first drum is 1000.0 kmol/h. We desire a liquid product from the first drum that is 30.0 mol% methanol ($x_1 = 0.300$). The second drum operates at a fraction vaporized of $(V/F)_2 = 0.250$. The equilibrium data are in Table 2-8 in Problem 2.D1.

a. Find the following for the first drum: y_1, T_1, $(V/F)_1$, and vapor flow rate V_1.

b. Find the following for the second drum: y_2, x_2, T_2, and vapor flow rate V_2.

a)



Material balance:

Overall MB: $F_1 = V_1 + L_1$

Component MB: $F_1 z_1 = V_1 y_1 + L_1 x_1$ $\rightarrow (1000)(0.55) = V_1(0.665) + (1000 - V_1)(0.300)$

$550 = V_1(0.665) + 300 - .300V_1$

$-300 \qquad -300$

$\frac{250}{.365} = \frac{.365 V_1}{.365}$

$$y_1 = -\frac{L_1}{V_1}x + \frac{F}{V_1}z_1$$

$X_1 = 0.665$

$\boxed{y_1 = 0.665 \quad T_1 = 78.0°C}$

$\boxed{V_1 = 684.9 \text{ kmol/h}}$

$x_1 = 0.665$

$\boxed{y_1 = 0.665 \quad T_1 = 78.0°C}$

$\boxed{V_1 = 684.9 \text{ kmol/h}}$

$\left(\frac{V}{F}\right)_1 = \frac{684.9}{1000} = \boxed{0.685}$

b) $(V/F)_2 = .250$

$F_2 = 684.9$ kmol/h

$z_2 = y_1 = 66.5\%$

$V_2 = 0.250(684.9) = \boxed{171.225 \text{ kmol/h}}$

Lever rule: $z_2 = \left(\frac{V}{F}\right)_2 y_2 + \left(\frac{1-V}{F}\right)_2 x_2$

$66.5 = 0.25y_2 + 0.75x_2$

trial: $y_2 = 77.9\%$

$0.25(77.9) + 0.75(50.0) = 19.5 + 37.5 = \cancel{57.0}$

$y_2 = 82.5\%$

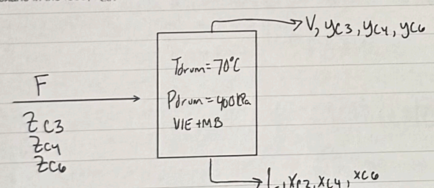
$0.25(82.5) + 0.75(60.0) = 20.6 + 45.0 = 65.6$

$\boxed{x_2 = 61.5\% \quad y_2 = 83.2\% \quad T_2 = 70.5°C}$

2) 15

D10. We have a mixture that is 35.0 mol% n-butane with unknown amounts of propane and n-hexane. We are able to operate a flash drum at 400.0 kPa and 70.0°C with $x_{C6} = 0.700$. Find the mole fraction of n-hexane in the feed, z_{C6}, and the value of V/F.

must equal to one



$T_{drum} = 70.0°C$

$P_{drum} = 400.0$ kPa

liquid mole fraction

using Chart...

Kvalue of n-butane @ 400kpa & 70°C $= 1.9$ K_{C4}

Kvalue of propane @ 400kpa & 70°C $= 5$ K_{C3}

Kvalue of n-hexane @ 400kpa & 70°C $= .3$ K_{C6}

using eqn. $\ln K = \frac{a_{T1}}{T^2} + \frac{a_{T2}}{T} + a_{T6} + a_{p1}\ln P + \frac{a_{p2}}{P^2} + \frac{a_{p3}}{P}$

$K_{C4} = 1.967$

$K_{C3} = 4.949$

$K_{C6} = 0.3225$

not too far from chart

$x_{C3} = 0.300 - x_4$

$x_{C6} = 0.700$

$x_{C4} = x_4$

$1 = \sum y_i = \sum (K_i x_i) = K_3(0.300 - x_4) + K_4(x_4) + K_6(0.700)$

$x_{C4} = 0.2383$ & $x_{C3} = 0.06174$

$y_{C3} = 4.949(0.06174) = 0.3055$

$y_{C4} = 1.967(0.2383) = 0.4687$

$y_{C6} = 0.3225(0.700) = 0.2258$

$\beta = \frac{V}{F}$

$z_{C4} = (1-\beta)x_{C4} + \beta y_{C4} \Rightarrow \beta = \frac{z_{C4} - x_{C4}}{y_{C4} - x_{C4}}$

$\beta = \frac{0.350 - 0.2383}{0.4687 - 0.2383} = 0.4849$

$\boxed{\frac{V}{F} = 0.485}$

$z_{C6} = \left(1 - \frac{V}{F}\right)x_{C6} + \frac{V}{F}y_{C6}$

$\boxed{\frac{V}{F} = 0.485}$

$z_{C6} = (1-\frac{V}{F})x_{C6} + \frac{V}{F}y_{C6}$

$(1-0.485)(.700) + (0.485)(0.2258)$

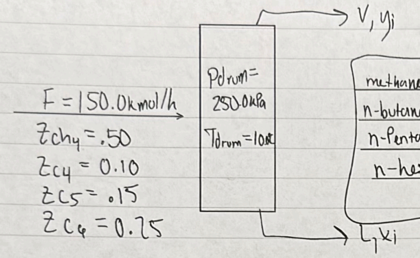
$0.3605 + 0.109513 = 0.470013$

$\boxed{z_{C6} = 0.470}$ 0.4089

3) 1.6

D16. A feed that is 50.0 mol% methane, 10.0 mol% n-butane, 15.0 mol% n-pentane, and 25.0 mol% n-hexane is flash distilled. $F = 150.0$ kmol/h, $p_{drum} = 250.0$ kPa, and $T_{drum} = 10.0°C$. Use DePriester charts or Eq. (2-28). Find V/F, x_i, y_i, V, and L.

$x_i = \frac{z_i}{1+\beta(K_i-1)}$



	x_i	y_i
methane	0.01614	0.83689
n-butane	0.12676	0.08137
n-Pentane	0.29168	0.05123
n-hexane	0.56525	0.03051

$\ln k_{zC6} = -6.84816+0+6.96783-3.03895 = -2.91928 = e^{-2.92198} = 0.05397$

$\ln k_{zCH4} = -1.12741+0+8.2445-3.21404+0.045519+0 = 3.9 = e^{3.9} = 51.861$

$\ln k_{zC4} = -4.92970+0+7.94986-3.46341 = -0.44 = e^{-.44} = 0.64195$

$\ln k_{zC5} = -5.87030+0+7.33129-3.20086 = -1.739 = e^{-1.73987} = 0.17554$

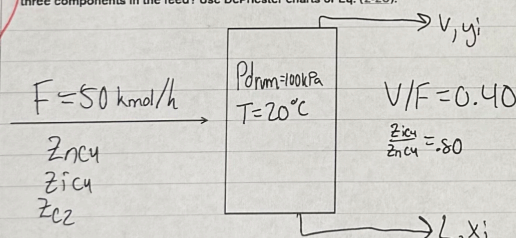
$\sum_i \frac{z_i(k_i-1)}{1+\frac{V}{F}(k_i-1)} = 0$ $\boxed{\frac{V}{F} = 0.58954}$ $\boxed{V = \beta F = 88.43 kmol/h}$

$L = (1-\beta)F = 61.57 kmol/h$

$\boxed{L = 61.57 kmol/h}$

4) 1

D15. We have a flash drum separating 50.0 kmol/h of a mixture of ethane, isobutane, and n-butane. The ratio of isobutane to n-butane is held constant at 0.80 (that is, $z_{iC4}/z_{nC4} = 0.80$). The mole fractions of all three components in the feed can change. If the flash drum operates at a pressure of 100.0 kPa, a temperature of 20.0°C, and V/F = 0.40, what are the mole fractions of all three components in the feed? Use DePriester charts or Eq. (2-28).



$V/F = 0.40$

$\frac{z_{iC4}}{z_{nC4}} = .80$

using R.R eqns. $\frac{V}{F} = 0.40$

$\sum_i \frac{z_i(k_i-1)}{1+\frac{V}{F}(k_i-1)} = 0$

ethane: $= \frac{3.813-1}{1+0.4(3.813-1)} = 1.2060$ $\quad K_{C2} = 3.813$

Isobutane: $\frac{0.303-1}{1+0.4(0.303-1)} = -1.0351$ $\quad K_{iC4}\ 0.303$

n-butane: $\frac{0.208-1}{1+0.4(0.208-1)} = -1.6796$ $\quad K_{nC4} = 0.208$

using chart

$z_{C_2} = 1 - 1.8a \quad z_{nC_4} = a$
$z_{iC_4} = 0.8a$

$(1 - 1.8a)(1.2060) = 1.2060 - 2.1708a$

$(0.8a)(-1.0351) = -0.8281a$

$1.2060 - (2.1708a + 0.8281a + 1.0796a) = 0$

$1.2060 - 4.0785a = 0$

$a = 0.2957$

$\boxed{z_{nC_4} = 0.2957}$ $\boxed{z_{iC_4} = 0.2366}$

$\boxed{z_{C_2} = 0.4677}$??